Lesson 2

Seven Stages of data Analysis

1.**Acquire**: Focuses on gathering data from its source.
2.**Parse**: Structures data to define meaningful components.
3.**Filter**: Eliminates irrelevant details to highlight key data.
4.**Mine**: Analyzes data to reveal underlying trends and insights.
5.**Represent**: Begins with a simple visual to identify complex relationships and patterns.
6.**Refine**: Enhances visual representation to transform data into clear information.
7.**Interact**: Shares analysis with stakeholders, enabling exploration and deeper understanding.

1. **Acquire (Gathering Data)**

* Collect raw data from primary or secondary sources (databases, APIs, sensors, logs).
* Ensure completeness and accuracy of data before moving forward.

2. **Parse (Structuring Data)**

* Organize raw data into structured formats like tables, rows, and columns.
* Define variables and attributes that will be meaningful for analysis.

3. **Filter (Selecting Relevant Data)**

* Remove noise, errors, or irrelevant records that don't serve the analysis goal.
* Focus on subsets of data that align with the intended question or problem.

4. **Mine (Analyzing Data)**

* Apply statistical, machine learning, or computational methods to extract patterns.
* Identify correlations, clusters, or anomalies hidden in the dataset.

5. **Represent (Initial Visualization)**

* Create basic charts or plots (scatter, line, bar) to see initial trends.
* Highlight potential relationships or structures in the data.

6. **Refine (Improving Visuals)**

* Enhance clarity by adjusting scales, colors, labels, and layouts.
* Transform rough visuals into polished, interpretable graphics.

7. **Interact (Exploring & Sharing)**

* Enable users to explore the data interactively (dashboards, filters, drill-downs).
* Share insights with stakeholders to support decision-making.

Principles of Chart Selection

For data visualization to be effective, charts must be :

- appropriate for the data and easily understandable. The main goal is to select a visualization that clearly communicates the key information and insights you want to conclude from your data.

Types of Charts

Charts can be broadly categorized based on the primary message they convey: Comparison, Distribution, Composition, and Relationship.

1. **Comparison Charts**
These charts are used to compare one or more variables to see differences or rank categories.
Use Cases: Comparing product revenues, tracking quarterly sales to find trends, or analyzing monthly footfalls in a store.
Common Chart Type: **Bar Chart**

Column Bar Chart: Best for comparing data across different categories or showing chronological data.

Horizontal Bar Chart: Ideal for comparing variables but can become cluttered with too many items.

Stacked Bar Chart: Used to compare data within a category to itself, often showing percentages.

2. **Distribution Charts**

Distribution charts help identify the normal tendency, range, and presence of outliers in a dataset.

Use Cases : Showing population distribution by age, student grades in an exam, or the heights of children in a class.
Common Chart Types: **Scatter Charts & Histograms**
These are effective for showing data distribution, clustering trends, and spotting anomalies. For instance, a histogram can show how many words are typically used in customer reviews.

3. **Composition Charts**
These charts illustrate how individual parts make up a whole, focusing on the contribution of each component.
Use Cases : Visualizing a country's population by religion, showing company market share, or breaking down sentiments in reviews (e.g., 69% negative, 18% neutral, 13% positive).
Common Chart Types: **Pie charts, stacked charts, and map-based graphs**.

4. **Relationship Charts**
Relationship charts are used to demonstrate the correlation between different variables, which can be positive, negative, or nonexistent.
Use Cases: Analyzing the relationship between marketing expenses and profit, or testing a hypothesis like "Does salary depend on IQ?".
Common Chart Types :

Scatterplot & Line Chart: Show relationships between variables.

Bubble Chart: Shows the relationship between words or concepts, where the bubble's size indicates frequency and color can represent another variable like sentiment.

Word Cloud: Represents the frequency of words in a text, with the size of the word being proportional to its usage.

Main Selection Question is : How can we ensure that the visualization communicates the message correctly to our audience?

Mistakes to avoid :

1. A chart with too much information will create clutter and discourage your audience from seeking insights.
2. Avoid putting a lot of data points in one chart rather use multiple visualizations to convey the story.
3. Always label the axes and provide chart headers. This informs our audience of the context of our analysis.
4. Depending on the data, horizontal, and vertical axes should be scaled appropriately.

Select the chart type wisely so that the audience can easily understand your analysis.

Students are Requested to go through each example pictorially to clearly understand the differences.